9.9 Seed Storage

Storage starts in the mother plant itself when it attains physiological maturity. Storage may be defined as the preservation of viable seeds from the time of collection until they are required for sowing. After harvesting the seeds are either stored in ware houses or in transit or in retail shop.

9.9.1. Classification of seeds based on storage

SEED STORAGE
Roberts (1973) classified seeds based on physiological behaviour during storing

ORTHODOX SEED	RECALCITRANT SEED
Seeds dried to low moisture of 5% (wet basis) and stored at low or Sub-freezing temperature for long period. Example:Cereals, pulses and oil seeds.	Seeds dried to high moisture of 20 - 50% (wet basis) and which cannot be successfully stored for long period. Example: Mango, Jack fruit, Coconut etc

SEED STORAGE
Ewart (1908) classified seeds into 3 categories based on life span or longitivity

Micro biotic:
Seed life span not exceeding 3 years.
Mesobiotic:
Seed life span not exceeding from 3 to 15 years.
Macrobiotic:
Seed life span not exceeding from 15 years to over 1000 years.

9.9.2. Methods of Seed Storage

i. Conventional Methods of Seed Storage

Conventional storage includes storage in Bamboo structure, mud and earthen structure, wooden structure and underground structure. In village level storage is done in large level in concrete/ cement silos, Metal or plastic drums and metal silos. Improved rural level storage structure includes storage in coal tar drum, udaipur bin, bamboo bin, pusa bin and metal bins.

ii. Modern Methods of Seed Storage

a. Seed storage in cryopreservation: It is the technique of germplasm conservation (storage of cells, tissue, embryo or seeds) by ultra-low temperature in liquid nitrogen at -196oC. It is not practical for commercial seed storage purpose, but is useful to store the valuable germplasm for use in future which cannot be preserved by conventional methods.

b. Seed storage in gene bank: In gene bank, seed storage is the preservation of seed under controlled environmental condition which will prolong the viability of the seeds for long periods. The temperature, relative humidity and seed moisture content. Containers and distribution arrangement vary for each and every type of seed.



Figure 9.13 Svalbard seed bank

c. Svalbard seed bank:
The seeds are stored in four ply sealed envelopes, and then placed into plastic tote containers on metal shelving racks. The storage rooms are kept at -18°C. The low temperature and limited access to O_2 will ensure low metabolic activity and delayed seed ageing. The permafrost surrounding will help to maintain low temperature of the seed when the electricity supply fails.



Nanotechnology in Agriculture:
Currently nanotechnology provides different nano devices and nano material that have a unique role in agriculture. For example Nano biosensors to detect moisture content and nutrient status in the soil. Nanotechnology can offer Nano-fertilizers for efficient nutrient management, Nano-herbicides for selective weed control in crop field, Nanonutrient particles to increase seed vigor, Nano-pesticides for efficient pest management. Hence, nanotechnology have greater role in crop production with environmental safety, ecological sustainability and economic stability.

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